

AETHERPOST

Technical Walkthrough

AI Powered Instagram Content Automation

Document Item	Details
Purpose	Operational, technical, and stakeholder handover
System	AetherPost
Automation	n8n
Source	Google Drive
AI	Caption and hashtag generation
Persistence	Supabase Storage + PostgreSQL
Publishing	Instagram Graph API
Audience	Stakeholders, operators, developers, maintainers
Status	Core automated publishing workflow operational

1. Executive Summary

AetherPost is an end-to-end automation system that converts content placed in a designated Google Drive folder into validated, AI assisted Instagram posts. The system is orchestrated through n8n and integrates Google Drive, Supabase, an AI content-generation service, and the Instagram Graph API.

The workflow reduces repetitive publishing work, separates supported images from unsupported files, prevents previously published images from being reposted, and maintains a persistent publication record.

A successful run retrieves files, splits them into individual items, filters for images, downloads and prepares a valid image, checks Supabase using the Google Drive file ID, generates and validates AI content, creates and publishes the Instagram media container, and logs the publication.

2. Business Purpose

- Reduce manual effort involved in routine Instagram publishing.
- Standardize publishing through a repeatable automation.
- Generate captions and hashtags with AI.
- Prevent accidental duplicate publication.
- Reject PDFs and other unsupported files before Instagram.
- Maintain a persistent audit trail of published content.

3. Scope

The implemented scope includes scheduled execution, Google Drive retrieval, multi-file processing, image filtering, image download and preparation, duplicate detection, AI content generation and validation, Supabase image storage, Instagram media creation and publication, and Supabase post logging.

Advanced analytics, approval queues, multi-account publishing, content calendars, and cross-platform publishing are potential future extensions rather than core implemented scope.

4. Solution Architecture

Layer	Component	Responsibility
Trigger	n8n Schedule Trigger	Starts the workflow.
Source	Google Drive + HTTP Request	Retrieves files.
Decision	n8n + JavaScript	Splits, filters, validates, and checks duplicates.
Persistence	Supabase	Stores images and publication records.
Publishing	Instagram Graph API	Creates and publishes media.
AI	AI provider	Generates caption and hashtag content.

5. End to End Workflow

Schedule Trigger → HTTP Request → Split Out → Filter Images → Google Drive Download → Prepare Image → Supabase Check Duplicate → New Image? → AI Generate Caption → Validate AI Output → Supabase Storage → Instagram Create Media Container → Wait for Processing → Instagram Publish → Supabase Log Post

Each file returned by Google Drive becomes an independent workflow item after Split Out, allowing mixed inputs to be evaluated separately.

6. Detailed Node Walkthrough

Schedule Trigger

Starts the automation on the configured schedule. It does not decide file type or duplicate status.

HTTP Request, Google Drive

Calls Google Drive and returns files[]. The query targets the configured folder and requests fields such as id, name, mimeType, createTime, and webViewLink. It must not filter out PDFs before the downstream image filter.

Split Out

Splits files[] into individual workflow items. One Drive file becomes one item.

Filter Images

Evaluates mimeType using an image/ prefix condition. TRUE continues to image processing; FALSE goes to Skip Non-Image.

Skip Non-Image

Terminal branch for PDFs, documents, and other unsupported files. It returns a skipped status and makes no Instagram request.

Google Drive, Download

Downloads the selected image as binary data. The expected binary file property must be available to the next stages.

Prepare Image

Prepares fileId, fileName, mimeType, public URL information, and other metadata.

Supabase, Check Duplicate

Queries public.post_logs using the current Google Drive fileId. The critical mapping is fileId → database column file_id.

New Image?

Zero matching records means new image and TRUE; an existing record means FALSE and routes to Skip Duplicate.

Skip Duplicate

Terminal branch that prevents a previously published image from being published again.

AI, Generate Caption

Generates caption and hashtag content for the new image.

Validate AI Output

Checks that the AI response is structurally valid before publication.

Supabase Storage, Upload

Stores the image and supplies a publicly reachable URL for Instagram.

Instagram, Create Media Container

Creates an Instagram media container using the public image URL and generated caption.

Wait for Processing

Allows the Instagram container to reach a publishable state.

Instagram, Publish

Publishes the successfully created media container.

Supabase, Log Post

Creates a post_logs record containing publication information and the Google Drive file ID.

7. Branch and Decision Logic

7.1 Non-Image Branch

A PDF or other unsupported file follows the FALSE output of Filter Images and terminates at Skip Non-Image. This is an intentional business outcome, not an error.

1. Drive returns the PDF.
2. Split Out creates a PDF item.
3. Filter Images evaluates its MIME type.
4. The condition is FALSE.
5. Skip Non-Image records a skipped status.
6. No image download, AI generation, or Instagram publication occurs.

7.2 Duplicate Branch

An image can remain in Drive after publication. AetherPost therefore does not treat presence in Drive as proof of newness. It compares the current Google Drive fileId with file_id stored in Supabase.

7. Prepare Image exposes fileId.
8. Supabase Check Duplicate queries post_logs.
9. A matching record means the file was already processed.
10. New Image? routes the item to Skip Duplicate.
11. No second Instagram post is created.

8. Supabase Data Model

Column	Purpose
id	Auto-generated identity / primary key.
image_name	Source filename.
image_url	Public image URL.

caption	Generated caption.
hashtags	Generated hashtags.
instagram_post_id	Instagram publication identifier.
status	Publication status.
error_message	Error information when applicable.
posted_at	Publication timestamp.
created_at	Record creation timestamp.
file_id	Google Drive file ID used for duplicate detection.

The id column is generated by the database. The workflow should not manually supply it.

9. Supabase Permissions

The REST API requires the service_role to have INSERT access to post_logs and appropriate access to its identity sequence. The operational grants used were:

```
GRANT INSERT ON TABLE public.post_logs TO service_role;
GRANT USAGE, SELECT ON SEQUENCE public.post_logs_id_seq TO service_role;
```

10. Instagram Integration

The publishing account is a professional Instagram account connected to the Meta developer application. The account was added as an Instagram Tester and the tester invitation was accepted before token generation.

The publishing sequence is: public image URL → media container creation → container ID → processing wait → media publication → Supabase logging. Access tokens are secrets and must never be committed to GitHub.

11. AI Content Generation

AI is responsible for generating social-media copy, while deterministic n8n logic controls routing, duplicate detection, validation, and publication. AI output is explicitly validated before it is passed to Instagram.

- Caption generation
- Hashtag generation
- Structured output
- Validation before publication

12. Security and Credential Management

The solution integrates multiple authenticated services. Credentials must be stored securely and separately from workflow logic.

- Never commit Instagram access tokens to GitHub.
- Never commit Supabase service-role keys to GitHub.
- Never commit AI API keys to GitHub.
- Never commit Google OAuth credentials to GitHub.
- Never commit Environment files containing secrets to GitHub.
- Use n8n credentials or secure environment configuration.

- Rotate credentials if accidentally exposed.

13. Operational Runbook

13.1 Normal New-Image Run

12. Place a new image in the monitored Drive folder.
13. Run the scheduled workflow or execute it manually for testing.
14. Confirm Google Drive returns the file.
15. Confirm Split Out creates an item.
16. Confirm Filter Images selects TRUE.
17. Confirm duplicate check returns no existing record.
18. Confirm AI output is valid.
19. Confirm Instagram media creation and publication succeed.
20. Confirm Supabase Log Post creates the record.

13.2 Non-Image Run

21. Place a PDF or other non-image file in the folder.
22. Run the workflow.
23. Confirm the file reaches Filter Images.
24. Confirm FALSE is selected.
25. Confirm Skip Non-Image executes.
26. Confirm no Instagram request is made.

13.3 Duplicate Run

27. Leave an already-published image in Drive.
28. Run the workflow again.
29. Confirm Supabase Check Duplicate finds its file_id.
30. Confirm New Image? selects FALSE.
31. Confirm Skip Duplicate executes.
32. Confirm no second Instagram post is created.

14. Acceptance Test Matrix

Test	Input	Expected Result	Acceptance
New image	New JPG/PNG	Publish and log	One new Instagram post and one log record.
Existing image	Previously published image	Skip duplicate	No new Instagram post.
PDF	PDF file	Skip non-image	No Instagram call.
Image + PDF	One of each	Independent branches	Image posts; PDF skips.
Invalid AI output	Malformed AI response	Validation blocks	No publication.
Unsupported ratio	Unsupported dimensions	Instagram rejects	Prepare image correctly before retry.
Successful logging	Published image	Log created	post_logs contains file_id and publication data.

15. Troubleshooting Guide

Symptom	Resolution
HTTP Request returns files: []	Verify Drive folder ID, query, permissions, and file location. Do not troubleshoot Split Out until files[] is populated.
Split Out returns 0 items	The files[] collection is empty; fix the Drive request first.
Filter Images does not execute	Inspect the specific execution and verify the previous node produced an item.
Skip Non-Image shows no data when opened alone	Run a complete workflow with a PDF; terminal branch nodes need an input item.
Duplicate image posts again	Verify duplicate query uses fileId and maps it to post_logs.file_id. Do not use a nonexistent field such as \$json.id when the item contains fileId.
Supabase permission denied	Verify INSERT permission on post_logs and identity-sequence permissions.
Invalid OAuth access token	Verify the complete token, correct account, and API flow; never add Bearer inside a query-string token value.
Aspect ratio not supported	Prepare the image to a supported Instagram feed aspect ratio before media-container creation.
Bucket not found	Verify the exact Supabase Storage bucket name in the upload endpoint.
Binary file data missing	Verify the Drive Download stage supplies binary data under the expected property.
AI returned invalid JSON	Inspect the AI response and validation/parser stage; enforce structured output before publication.

16. Testing Strategy

The workflow was validated through new-image, duplicate-image, PDF, and mixed-file scenarios. A mixed test is especially important because the same Google Drive execution can contain both supported and unsupported files.

Scenario	Expected Path
New image	Image filter → duplicate check → AI → Instagram → log
Existing image	Image filter → duplicate check → Skip Duplicate
PDF	Image filter → Skip Non-Image
Image + PDF	Image follows publishing path; PDF follows skip path
Existing image + new PDF	Existing image skips; PDF skips; no new post

17. Maintenance and Ownership

Area	Maintenance Responsibility
n8n	Workflow, schedule, credentials, execution history, node changes.
Google Drive	Source folder, file organization, access permissions.
Supabase	Storage bucket, post_logs schema, permissions, retention.
Meta / Instagram	Professional account, app configuration, tokens, API changes.
AI provider	API key, model availability, response format, usage limits.
GitHub	Version control and documentation; never store secrets.

18. Change Management Recommendations

- Export or back up the working n8n workflow before changing production logic.

- Test changes with a dedicated test image and test Instagram account.
- Test duplicate protection whenever file-identification logic changes.
- Test image-only, PDF-only, and mixed-file scenarios whenever the Drive query changes.
- Validate Meta/Instagram API changes against current provider documentation before changing endpoints.
- Record Supabase schema and permission changes as deployment changes.
- Rotate credentials when necessary and update secure n8n configuration.

19. Deployment / Handover Checklist

- n8n workflow imported and saved.
- Schedule configured.
- Google Drive credentials connected.
- Correct Drive folder configured.
- HTTP Request returns files[].
- Split Out configured for files.
- Filter Images configured for image MIME types.
- Skip Non-Image connected.
- Google Drive Download provides binary image data.
- Prepare Image provides fileId and required metadata.
- Supabase Storage bucket exists.
- post_logs table exists.
- service_role has required database permissions.
- Duplicate query maps fileId to file_id.
- New Image? routes existing files to Skip Duplicate.
- AI credentials configured.
- AI output validation configured.
- Instagram professional account connected.
- Instagram tester role accepted where required.
- Instagram access token generated and stored securely.
- Instagram media creation tested.
- Instagram publication tested.
- Supabase logging tested.
- New-image, duplicate-image, PDF, and mixed-file scenarios tested.
- No credentials committed to GitHub.

20. Current System State

The core AetherPost publishing pipeline is operational. New images can move from Google Drive through filtering, duplicate detection, AI content generation, Instagram media creation, Instagram publication, and Supabase logging. PDFs and other non-image inputs are routed to the skip branch. Previously published images are checked against Supabase using their Google Drive file ID.

The system depends on external services remaining available and compatible with the configured API versions, credentials, permissions, and account settings.

21. Stakeholder Summary

AetherPost turns a simple content-upload action into a controlled automated publishing process. Stakeholders can place content in the configured Drive location while the system handles file classification, duplicate prevention, AI assisted copy generation, Instagram publication, and publication logging.

The modular architecture separates content storage, AI generation, persistent state, and social-media publication. This provides clear maintenance boundaries and creates a foundation for future capabilities such as approvals, scheduling, analytics, multiple accounts, and additional social platforms.

Appendix A Quick Reference

Requirement	Expected Behavior
New image	Publish and log.
Previously published image	Skip Duplicate.
PDF/document	Skip Non-Image.
Multiple files	Process independently.
Invalid AI response	Stop before Instagram.
Unsupported image ratio	Correct image preparation before retry.
Successful post	Create post_logs record.
Future execution	Use file_id to prevent duplicate publication.

Appendix B Key Terms

Term	Meaning
fileId	Google Drive identifier for the source file.
file_id	Supabase post_logs column storing the source identifier.
files[]	Google Drive API file collection.
Media Container	Instagram object created before final publication.
publicUrl	Public Supabase Storage image URL used by Instagram.
post_logs	Supabase publication history and duplicate-detection table.
Skip Non-Image	Terminal branch for unsupported file types.
Skip Duplicate	Terminal branch for previously published images.